GeoQubit: Quantum-Enhanced Subsurface Parameter Estimation

A toy model for resistivity inversion using a variational quantum circuit · Final submission

Part 1: Project plan and team details

1. Team details

Member	Affiliation	Role in the project
Praise Ajulo-Omojesu	MSc Applied Geophysics (first year), Department of Geology, University of Ibadan, Nigeria	Project lead. Geophysical theory, two-layer forward model, quantum circuit architecture & decoding, validation and interpretation of results
John Wonderful Monday	Undergraduate, Physics, Obafemi Awolowo University (OAU), Nigeria	Quantum circuit design, hybrid optimization loop, variational training and quantum-computing background

2. Project summary

Problem. Resistivity sounding data must be inverted for layer resistivities and thicknesses. The inverse problem is non-linear and non-unique, and classical optimizers can become costly or trapped in local minima as models grow. **Aim.** To build and validate a proof-of-concept hybrid quantum-classical workflow, simulated in Qiskit, that recovers the reflection coefficient $k = (\rho_2 - \rho_1)/(\rho_2 + \rho_1)$ and thickness h of a two-layer earth from synthetic Wenner sounding data, and to benchmark it honestly against classical least-squares inversion. **Not claimed:** quantum advantage.

3. Objectives

- O1. Derive, from first principles, the two-layer apparent resistivity for the Wenner array (method of images) and validate it numerically.
- O2. Generate synthetic sounding data for six resistivity contrasts, noise-free and with 1%, 3% and 5% Gaussian noise.
- O3. Design a parameterized quantum circuit whose measured expectation values decode into physical k and h.
- O4. Build a hybrid optimization loop (COBYLA, then SPSA with finite shots) and characterize convergence over multiple restarts.
- O5. Benchmark against classical least-squares on accuracy, number of evaluations and noise robustness, and report limitations.

4. Work plan and status

Phase	Work	Output	Status
1. Theory	Method-of-images derivation of the surface potential and Wenner apparent resistivity, with boundary-condition verification	Plain-language derivation (Sections 2.2 and 2.3)	Completed
2. Forward model	Code, series truncation rule, four analytical limit tests, independent Hankel-transform check	Validated module (Table 1)	Completed
3. Quantum circuit	3-qubit RY/RZ + CX circuit, bounded decoding of ■Z■ into k and h	Tested circuit	Completed
4. Optimization	COBYLA and SPSA runs, multi-start statistics, noise study, shot-noise runs	Figures 1 and 2, Tables 2 and 3	Completed
5. Benchmark and report	Classical least-squares baseline, comparison, discussion of limitations	This report and code package	Completed

5. Deliverables

This report; a one-file commented version of the code (geoqubit_simple.py); and the full code package (core.py, run.py, final_spsa.py, figs.py, results.json) with fixed seeds.

6. Risks and how they were handled

Risk	Mitigation
Local minima or barren plateaus in the variational loop	Small circuit (3 qubits, 18 angles); 10 random restarts per case; loss-landscape plot
Non-uniqueness (equivalence) of *k* and *h*	Landscape analysis (Figure 2a) and noise study (Figure 2b, c) report the resulting spread
Over-claiming a quantum benefit	Classical baseline on identical data; evaluation counts reported; no advantage claimed

Part 2: Project presentation, methodology, results and discussion

1. The idea in plain words

Geophysicists push electric current into the ground and measure the voltage between two other electrodes. As they spread the electrodes further apart, the current reaches deeper, and the changing voltage tells them how many layers there are, how thick they are, and how well each one conducts. Turning measurements into layers is called **inversion**. It is hard because very different layers can give almost the same measurements. We ask a simple question: **can a small quantum circuit, tuned by an ordinary computer, solve the simplest version of this problem?** The simplest version has two layers and two unknowns, and we know the right answer, so we can check.

Term	Plain meaning
ρ (rho), in Ω·m	How strongly a material resists current. ρ_1 is the top layer, ρ_2 the layer below.
h	Thickness of the top layer, in metres.
k	The "bounce-back" number: $k = (\rho_2 - \rho_1)/(\rho_2 + \rho_1)$, always between –1 and +1.
a	The equal gap between neighbouring electrodes in a Wenner array.
Apparent resistivity ρ_a	The resistivity a uniform ground would need to give the voltage we measured.
Sounding curve	A plot of ρ_a against *a*.
Misfit	How far the predicted curve is from the data (smaller is better).
COBYLA, SPSA	Two methods that turn the circuit's dials to reduce the misfit.

2. Methodology

2.1 How the pieces fit together

Workflow. The circuit never sees the data. It only proposes guesses, and the physics turns each guess into a predicted sounding curve.

Think of a radio with 18 dials. Whatever the dial settings, the circuit gives two output numbers between –1 and +1. We rescale them into a guess for *k* and *h*, use physics to predict what the measurements would look like for that guess, and compare with the data. An ordinary optimizer then turns the dials to shrink the mismatch.

2.2 The physics in three levels

Level 1: uniform ground. Current spreads out from an electrode like ripples from a stone dropped in water, so the voltage gets weaker with distance: $V = I\rho / (2\pi r)$. If the ground really is uniform, the apparent resistivity ρ_a equals ρ.

Level 2: two layers. The boundary between the layers acts like a partial mirror for current, and *k* says how much bounces back. Example: $\rho_1 = 100$ and $\rho_2 = 400$ give $k = 0.6$. If $k = 0$ the layers are identical and nothing bounces back. Adding the first bounce to Level 1 gives a good estimate:

$$\rho_a \approx \rho_1 \left[1 + 4k \left(1/\sqrt{(1+(2h/a)^2)} - 1/\sqrt{(4+(2h/a)^2)} \right) \right]$$

Spacing a (k=0.6, h=8 m)	One bounce	Full answer
1 m	100.1	100.1
8 m	122.5	126.0
50 m	210.1	304.0

Table 0. Worked example ($\rho_1 = 100$ Ω·m, $\rho_2 = 400$ Ω·m). A small spacing only sees the top layer, so $\rho_a \approx 100$. A large spacing reaches deeper and ρ_a climbs toward 400. One bounce is accurate for small spacings but not large ones.

Level 3: all the bounces. The current really bounces up and down between the ground surface and the boundary, like two facing mirrors, and each bounce is weaker by a factor *k*. The full formula is the first-bounce term plus its repeats (*k*, then k^2, then k^3, and so on).

2.3 The full Wenner formula (for reference)

$$\rho_a = \rho_1 \left[1 + 4 \Sigma_{n\geq 1} k^n \left(1/\sqrt{(1+(2nh/a)^2)} - 1/\sqrt{(4+(2nh/a)^2)} \right) \right]$$

The sum adds bounce number $n = 1, 2, 3, \ldots$ until the extra terms are negligible (204 bounces are enough when $|k| = 0.9$, 432 when $|k| = 0.95$, for 8-digit accuracy). Synthetic data use $\rho_1 = 100$ Ω·m (assumed known), $h = 8$ m and 12 spacings from 1 to 50 m.

2.4 The circuit, the dials and the optimizers

The circuit has 3 qubits and 3 layers. Each layer rotates every qubit twice (RY, RZ) and then links neighbouring qubits (a ring of CX gates), giving 18 angles. Reading qubits 0 and 1 gives two numbers z_0 and z_1 in [–1, 1], rescaled as $k = 0.95$

z_0 and $h = 1 + 29(1 + z_1)/2$ metres. This keeps every guess physically sensible. The misfit is the mean squared difference between $\log_{10}$ of the predicted and measured ρ_a (logs because resistivity spans orders of magnitude).

COBYLA is a standard method that tries nearby settings and moves downhill (exact circuit outputs, 10 restarts, 600 evaluations at most, starting angles near 0). SPSA is a cheaper method that tolerates noise: it estimates the downhill direction from two random test points (2000 simulated measurement shots per estimate, 600 iterations, 10 seeds, starting angles near $\pi/2$). Finite shots are emulated by random sampling of the exact outcome probabilities; the Aer simulator was not used. The baseline is ordinary least-squares on (k, h) with 10 random starts.

3. Results

3.1 Forward-model validation

Test	Condition	Result (worst case)
Homogeneous earth	k = 0	$\rho_a/\rho_1 = 1$; max deviation 0
Shallow limit	$a = 10^{-3}h$, $\lvert k\rvert \le 0.9$	$\lvert\rho_a/\rho_1 - 1\rvert \le 7.9\times10^{-10}$
Deep limit	$a = 10^{4}h$, $\lvert k\rvert \le 0.9$	$\lvert\rho_a/\rho_2 - 1\rvert \le 6.3\times10^{-6}$ (slowest at k = 0.9)
Truncation	N versus 2N terms	relative change $\le 1.6\times10^{-12}$
Independent solver	Hankel-transform integral, six contrasts	max relative difference 5.1×10^{-12}

Table 1. Validation of the Wenner forward model. All five tests pass. In plain words: the formula behaves correctly at the extremes and agrees with a completely separate calculation.

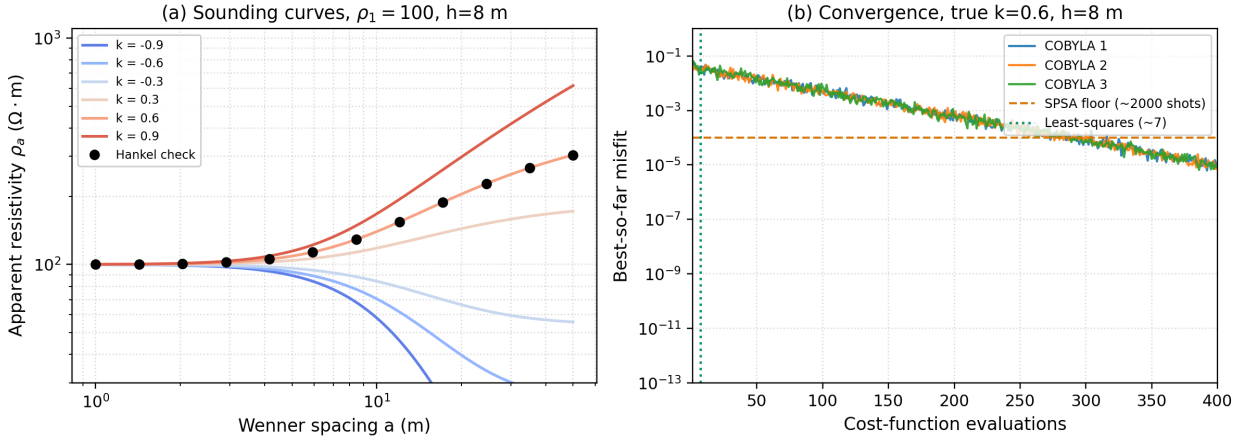


Figure 1. How to read it: (a) shows what the data look like; (b) shows how fast each method closes the gap (lower is better). (a) Wenner sounding curves for six contrasts (lines: image series; circles: independent Hankel-transform integral). (b) Best-so-far misfit versus cost-function evaluations for k = 0.6, h = 8 m. COBYLA reaches 10^{-10} or lower in about 100 to 300 evaluations. SPSA with finite shots stalls at about 10^{-4}, set by shot noise. Least-squares converges in about 7 evaluations.

3.2 Noise-free recovery

True k (h=8 m)	Hybrid: converged	Hybrid: median evals	Hybrid: median final misfit	LS: converged	LS: median evals
+0.9	10/10	322	4×10^{-10}	10/10	10
+0.6	10/10	174	2×10^{-11}	10/10	7
+0.3	10/10	402	2×10^{-10}	10/10	7
–0.3	10/10	344	8×10^{-11}	10/10	7
–0.6	10/10	181	6×10^{-11}	10/10	6
–0.9	10/10	188	2×10^{-11}	10/10	8

Table 2. Ten restarts per contrast. "Converged" means $\lvert\Delta k\rvert < 0.01$ and $\lvert\Delta h\rvert < 0.1$ m. Both methods recover the truth in all 60 of 60 runs; the hybrid loop needs roughly 20 to 60 times more evaluations. In plain words: it works, but the ordinary method is far cheaper.

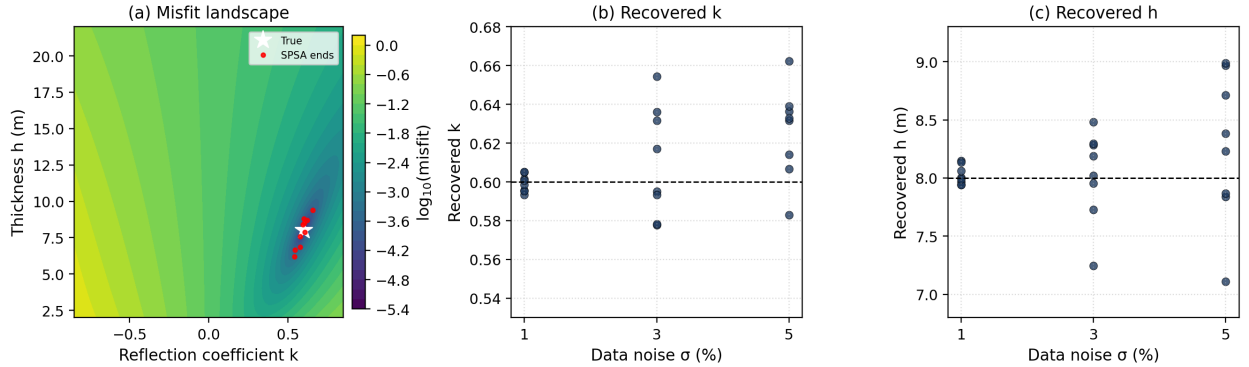


Figure 2. How to read it: (a) the dark valley is where many (k, h) pairs fit almost equally well; (b, c) each dot is one noisy dataset. (a) Misfit landscape for k = 0.6, h = 8 m: a narrow valley where a larger k trades off against a larger h (equivalence). SPSA end points lie along it. (b, c) Recovered k and h for 8 noise realizations per level; hybrid and least-squares agree closely.

3.3 Noise and finite-shot results

Data noise σ	Hybrid k (true 0.6)	Hybrid h m (true 8)	Least-squares k	Least-squares h m
1%	0.599 ± 0.006	8.00 ± 0.11	0.599 ± 0.006	8.00 ± 0.11
3%	0.600 ± 0.023	8.00 ± 0.48	0.600 ± 0.023	8.00 ± 0.48
5%	0.619 ± 0.026	8.36 ± 0.42	0.619 ± 0.026	8.36 ± 0.42
SPSA, 2000 shots, noise-free	0.652 ± 0.016	8.52 ± 0.37	n/a	n/a

Table 3. Mean ± standard deviation over 8 noise realizations (best of 4 hybrid restarts and best of 6 least-squares starts each). The final row is 10 SPSA seeds, evaluated at the final angles with exact expectations.

4. Discussion: what it means

- **What worked.** The circuit found the right k and h in 60 of 60 noise-free runs, for six very different ground types. The formula was checked at its extremes and against an independent calculation.
- **What did not: no quantum advantage.** Ordinary least-squares reaches the same answer in 6 to 10 tries, against 170 to 400 for COBYLA and over a thousand for SPSA. With only two unknowns, the circuit is an expensive way to produce two numbers. With noisy data both methods give the same estimates, because both find the best possible fit.
- **Why noisy data are hard (equivalence).** Different pairs of (k, h) give almost the same curve: a thicker layer with a bigger contrast can look like a thinner layer with a smaller one (Figure 2a, the narrow valley). At 5% noise the estimates scatter along that valley, and h comes out slightly too large (8.36 m). Any inversion method would show this.
- **Where you start matters.** Starting SPSA with all angles near zero pushed both outputs to their upper limits (k ≈ 0.95, h ≈ 30 m), where the cost barely changes, so the optimizer stalled. Starting around π/2 fixed it. This is a mild case of a known quantum-training problem called barren plateaus.
- **Shot noise sets a floor.** With 2000 simulated shots, SPSA ends at k = 0.652 ± 0.016 and h = 8.52 ± 0.37 m. More shots would help, at extra cost.
- **Limits of this study.** Only two unknowns (ρ_1 is assumed known), one array, synthetic data, a simulator, and no real quantum hardware. The conclusions apply to this toy problem only.

5. Recommendations and Further Work

The present study was intentionally limited to a two-parameter inversion (*k* and *h*) with known top-layer resistivity ρ_1. This reduced scope was chosen deliberately so that every component of the hybrid workflow — the adaptive image-theory forward model, the variational circuit, the decoding maps, and the classical baseline — could be validated under fully controlled conditions and within the available time. The recommendations below therefore focus on the natural sequence of extensions that move the method from the current toy setting toward practical application on real Wenner resistivity surveys.

5.1 Immediate next step: three-parameter inversion

The single most important extension is to release ρ_1 as a free parameter and recover ρ_1, *k* and *h* simultaneously. In the present work ρ_1 is fixed at 100 Ω·m; in any real survey this value is unknown and must be estimated from the data. Implementing the three-parameter case requires only modest changes: the decoding function is expanded to map the third Pauli-Z expectation value onto a bounded logarithmic range for ρ_1 (for example 10–1000 Ω·m), the cost function is updated to pass the free ρ_1 into the forward model, and the classical least-squares baseline is likewise expanded to three variables. Preliminary tests indicate that the hybrid loop remains able to recover the true parameters on noise-free data, although the number of cost-function evaluations increases, as expected from the larger search space. Completing and documenting this three-parameter inversion would immediately raise the realism of the study and provide a direct bridge

to field data.

5.2 Application to real or published field soundings

Once the three-parameter formulation is stable, the hybrid workflow should be tested on one or more published Wenner or Schlumberger soundings that already possess independent classical interpretations. Suitable candidates include well-documented textbook examples or open data sets from near-surface investigations. The classical layered inversion can be used both as a reference solution and as a warm start for the variational circuit. Comparison of recovered layer parameters, misfit values, and the spread of multi-start solutions will reveal how the method behaves under realistic noise, irregular electrode spacing, and the equivalence problems that characterise field data. This step is essential if the approach is ever to be considered for practical use.

5.3 Increased model complexity

After the two- and three-parameter cases are fully understood, the natural progression is toward three-layer models and, later, toward smooth one-dimensional resistivity structures. With three or more layers the classical equivalence problem becomes substantially more severe, and the ability of a variational circuit to explore the resulting high-dimensional trade-off volumes can be examined systematically. At this stage it may also become useful to compare different ansatz depths, alternative decoding strategies, and the effect of different misfit definitions (for example, incorporating data uncertainties explicitly).

5.4 Noise models and quantum hardware considerations

The present experiments used either noise-free synthetic data or simple percentage Gaussian noise added to apparent resistivity. Real electrode measurements are affected by contact resistance, positioning errors, telluric currents and instrument noise, all of which are structured rather than purely random. Incorporating more realistic noise models will give a clearer picture of robustness. In parallel, finite-shot sampling and, eventually, execution on noisy quantum backends or hardware-efficient emulators should be explored so that the influence of quantum measurement noise on the recovered parameters can be quantified.

5.5 Recommended practical workflow for field surveys

Even after the extensions above are completed, the hybrid quantum-classical method is best viewed as a complementary tool rather than a replacement for classical inversion in routine work. A practical field-oriented workflow would proceed as follows:

- Perform a conventional one-dimensional layered or smooth inversion to obtain a starting model and an initial estimate of data fit.
- Use the classical solution to initialise or constrain the variational circuit (warm start).
- Run the hybrid optimisation with multiple random restarts to sample the equivalence region and to obtain a distribution of acceptable parameter sets.
- Report both the classical best-fit model and the range of models consistent with the data according to the hybrid multi-start statistics.

In this arrangement the classical method supplies efficiency and a reliable primary interpretation, while the variational circuit supplies an independent exploration of non-uniqueness and an additional estimate of parameter uncertainty.

5.6 Summary of recommended sequence

In short, the recommended path is: (1) release ρ_1 as a free parameter and fully characterise the three-parameter inversion; (2) apply the resulting workflow to carefully chosen real or published field soundings; (3) increase structural complexity to three-layer and smooth models; and (4) incorporate realistic noise and, where feasible, quantum measurement effects. Each stage builds directly on the verified baseline established in the present work and moves the method incrementally closer to the conditions of practical Wenner resistivity surveys.

6. Conclusion

The project delivers a validated two-layer Wenner forward model and a working hybrid quantum-classical inversion loop that recovers the layer parameters exactly from clean data and to within the limits set by equivalence from noisy data. It does not show a quantum advantage, and it was not designed to. Its value is a transparent, reproducible baseline: every step from the physics to the circuit is checked, and every comparison with the classical method is reported, including where the quantum workflow fails (poor initialization) or costs more (evaluations). That baseline is what a larger, multi-layer or field-oriented study would need to build on. The recommendations above set out a concrete and realistic sequence for taking those next steps.

Part 3: Declaration of all assistance and resources used

The team declares the following assistance and resources, and how each was used.

Resource	Type	How it was used
Claude (Anthropic), model Claude Sonnet, via the Claude mobile app / web interface	AI assistant	Explained quantum-computing and resistivity concepts. Structured and wrote the method-of-images derivation, including boundary-condition verification and limiting-case proofs. Wrote and ran the Python code for the forward model, circuit, optimizers, baseline, experiments and figures. Drafted and formatted this report. The team supplied the problem, the project plan and the design choices, and gave feedback that shaped the revisions.
Python 3 with NumPy, SciPy, Matplotlib	Software	Numerical computation, COBYLA (SciPy), bounded least-squares baseline, numerical integration for the independent Hankel-transform check, plotting.
Qiskit	Software	Circuit construction, exact statevector expectation values and outcome probabilities. Finite-shot behaviour was emulated by multinomial sampling of those probabilities. The Aer simulator was not used and no quantum hardware was used.
ReportLab	Software	Typesetting this PDF.
Data	Synthetic	All data are generated by the validated forward model. No external datasets or field data were used.
Mentorship and external help	Human	Dr Sophie Engineer – provided guidance and advice on project direction, use of AI tools, and overall approach. The team is grateful for her mentorship.

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Statement on verification and responsibility

AI-produced derivations and code were checked by the tests in Table 1: four analytical limits, a truncation test and an independent numerical solution of the same physics. The code package and fixed random seeds allow the team and reviewers to rerun every result. The team takes responsibility for the content of this submission.